Temperature Effects on LLM Behavior

1. Comparative Table of Models & Temperatures

Model	Temp 0.10	Temp 0.32	Temp 0.50	Temp 0.55	Temp 0.70	Temp 0.78/0.85	Temp 1.00	Temp 1.30	Temp 1.50
GPT-2	Repetitive "Yes, Earth is flat."	Mostly "Earth is not flat"	Repeats "Earth is flat"	Adds nonsensical facts (radius, divisions)	Longer, contradictory explanations (layers, water)	Surreal (horses, polar bears)	Imaginative, cosmic ideas	—	—
DistilGPT-2	Very repetitive, one cluster	Still repetitive	Locked into one pattern	Slight variation, alternating fragments	More clusters, longer explanations	Odd phrasing, more variety	Highly diverse, multiple clusters	—	—
GPT-2 XL	Loops: "No. Earth is round."	Same loop	One short deterministic response	Many clusters, explanatory fragments	Multi-cluster, longer but contradictory	Longer with incorrect facts, drifting	Speculative, abstract, less logical	Rambling, confused analogies	Disorganized, philosophical tangents
GPT-Neo 1.3B	Starts "No," then loops with "Yes / Earth is sphere"	Single block "No, not flat"	Short looping "No/Because…"	Emergent variety but wrong facts (diameter 180 miles)	Flips between flat vs sphere, weak coherence	Contradictory astronomy claims, meandering	Meta/tangential, inconsistent	Rambling, analogies	Disorganized, conspiracy-like rambles

2. Key Observations & Insights

Patterns Across All Models

- **Low Temperatures (0.10–0.32)** → Very deterministic, repetitive, "safe" outputs. Models latch onto a single phrase and repeat it with little creativity.
- **Mid Temperatures (0.50–0.70)** → Creativity emerges but coherence weakens. Some models hallucinate details (e.g., fake numbers, "layers of Earth").
- **High Temperatures (0.85–1.50)** → Outputs become highly varied, imaginative, but often factually incorrect, contradictory, or nonsensical.

Model-Specific Traits

- **GPT-2**: Quickly shifts from repetitive denial to surreal, creative imagery at mid-high temperatures. Strong jump in imagination around **0.78–1.00**.
- **DistilGPT-2**: Shows **less variety overall**, with smaller jumps in creativity compared to GPT-2. More constrained but also less chaotic.
- **GPT-2 XL**: Diversity Handles better than smaller models but loses coherence as temp rises. At high temps, it produces rambling, abstract text.
- **GPT-Neo**: Prone to contradictions (flat vs sphere in same response). At high temps, tends toward conspiracy-like or rambling narratives.

Interesting Facts

1. **Threshold Behavior**: Many models show a *sharp change* between **0.55–0.70** — moving from repetitive answers into contradictory or creative zones.
2. **Size ≠ Stability**: Larger model (GPT-2 XL) doesn't always mean more coherent — it becomes more verbose but loses logical consistency at high temps.
3. **Hallucination Examples**:
    - GPT-2 invented "radius" and "divisions" at Temp 0.55.
    - GPT-Neo confidently claimed Earth's diameter ≈ *180 miles* at Temp 0.55.
4. **Surrealism Emerges at Mid-Temps**:
    - GPT-2 introduced *polar bears and horses* while discussing Earth's shape.
    - GPT-Neo added *confused analogies and tangents*.
5. **Meta Behavior**: At Temp 1.00, GPT-Neo and GPT-2 XL sometimes step into self-referential, abstract, or philosophical commentary, a sign of drifting away from prompt grounding.

3. Conclusion

- **Temperature is a critical control**: Low values enforce safe, repetitive answers; high values unlock creativity but at the cost of coherence and truth.
- **Optimal tradeoff**: Around **0.55–0.70** offers the richest “interesting” content (variety without complete loss of sense), though accuracy is still fragile.
- **Model personalities** emerge: GPT-2 is more surreal, DistilGPT-2 is conservative, GPT-2 XL is verbose but unstable, and GPT-Neo is contradictory and rambling.